

作 业

1. Consider the following relational schema. An employee can work in more than one department; the pct_time field of the Works relation shows the percentage of time that a given employee works in a given department.

Emp(eno, ename, age, salary)

Works (eno, deptid, pct_time)

Dept (deptid, deptname, budget, manageid)

- (1) Give a SQL statement to find the employees with salaries over 50,000, ages under 40 and working hours in software development departments over 50% , list their NOs and *names*.

- (2) For the SQL statement in (1), give an optimized query tree.

2. Consider the following tables mentioned above in Question 2

factory(factoryID, name, manager, account)

employee(employeeID, name, age, factoryID)

airconditioner(serialID, date, model, price, factoryID)

storehouse(houseID, size, address)

stored(serialID, houseID)

order(orderID, customerID, model, num_ordered, factoryID)

customer(customerID, name, address)

- (1) Give a SQL statement to find the air conditioner model. It is required that the number of the ordered products of this model is not below 200, the model's price is higher than 2000, and all ordered products of this model are stored in the storehouse with *houseID* BJ101. List the model, the number of the ordered product. (4 points)

- (2) For the SQL statement in (1), give an optimized query tree.

作业答案及例题

1. Consider the following relational schema. An employee can work in more than one department; the `pct_time` field of the `Works` relation shows the percentage of time that a given employee works in a given department.

`Emp`(`eno`, `ename`, `age`, `salary`)

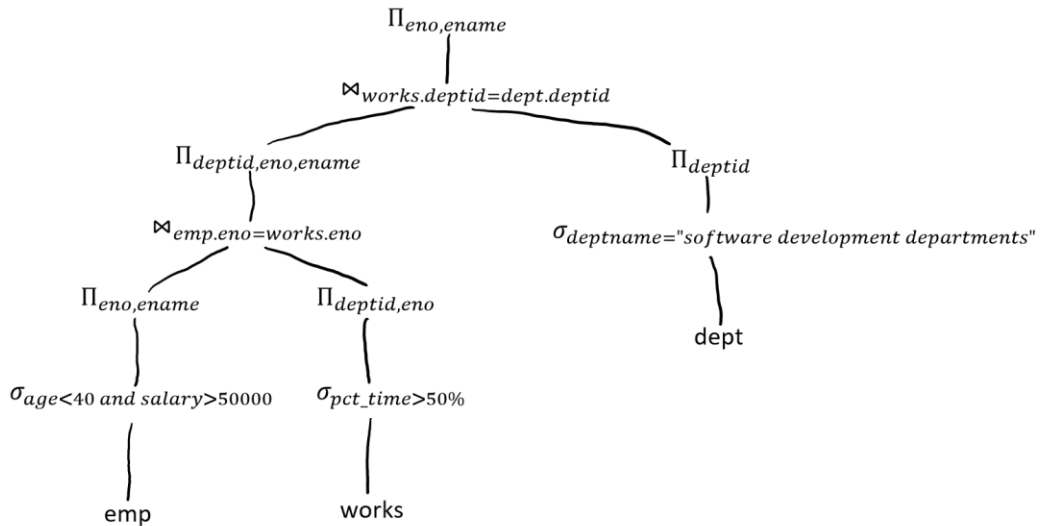
`Works` (`eno`, `deptid`, `pct_time`)

`Dept` (`deptid`, `deptname`, `budget`, `manageid`)

- (1) Give a SQL statement to find the employees with salaries over 50,000, ages under 40 and working hours in software development departments over 50% , list their NOs and *names*. (3 points)

```
select Emp.eno, ename
from emp, works, Dept
where emp.eno=works.eno and works.deptid=dept.deptid
      and salary>50000 and age<40 and pct_time>50
      and deptname="software development"
```

- (2) For the SQL statement in (1), give an optimized query tree.



2. (10 points) Consider the following tables mentioned above in Question 2

factory(factoryID, name, manager, account)

employee(employeeID, name, age, factoryID)

airconditioner(serialID, date, model, price, factoryID)

storehouse(houseID, size, address)

stored(serialID, houseID)

order(orderID, customerID, model, num_ordered, factoryID)

customer(customerID, name, address)

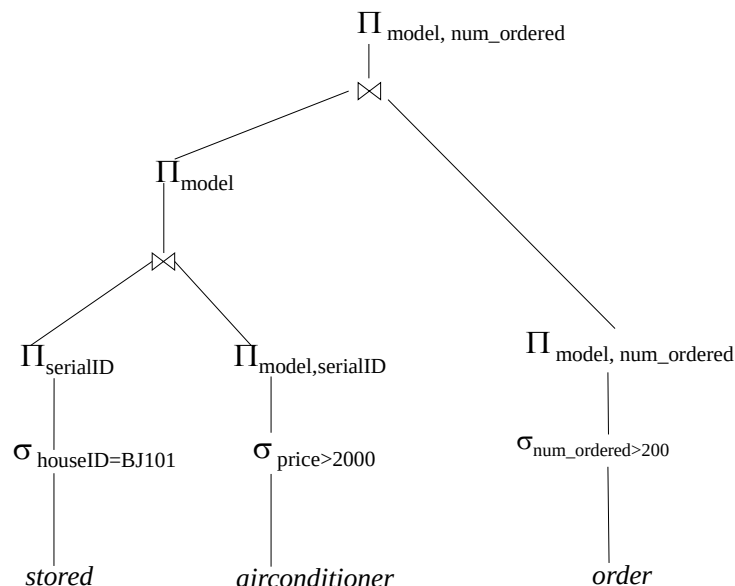
(1) Give a SQL statement to find the air conditioner model. It is required that the number of the ordered products of this model is not below 200, the model's price is higher than 2000, and all ordered products of this model are stored in the storehouse with *houseID* BJ101. List the model, the number of the ordered product. (4 points)

```
select  model, num_ordered
from    stored natural join airconditioner natural join order
where   houseID=BJ101 and price>2000 and num_ordered>200
```

说明：3 张表、2 个连接条件/自然连接、3 个选择条件；

以在 from 子句中不使用自然连接，改为在 where 子句中写出连接条件.

(2) For the SQL statement in (1), give an optimized query tree.



3. Consider the following relations in a Library database, where the primary keys are underlined.

Books (BookIndexNo, BookTitle, author, bookno, PublishUnit, price, year-published),

Readers (CardID, student-name, student-school, sex),

Loans(CardID, BookIndexNo, borrowing-date, priority).

For the query “Find out the names of those students in School of Computer Science, who borrow the book “Database System Concepts” between 2015/03/01 and 2015/06/20, and these borrowed books were published in the year of 2012”

(1) Give an SQL statement for this query.

(2) For the query in (1), convert it into an optimized query tree by means of heuristic optimization.

Answers:

(1)

Select studentname

From Books, Readers, Loans

Where Books.BookIndexNo= Loans.BookIndexNo and

Loans.CardID= Readers.CarID and

student-school= School of Computer Science and

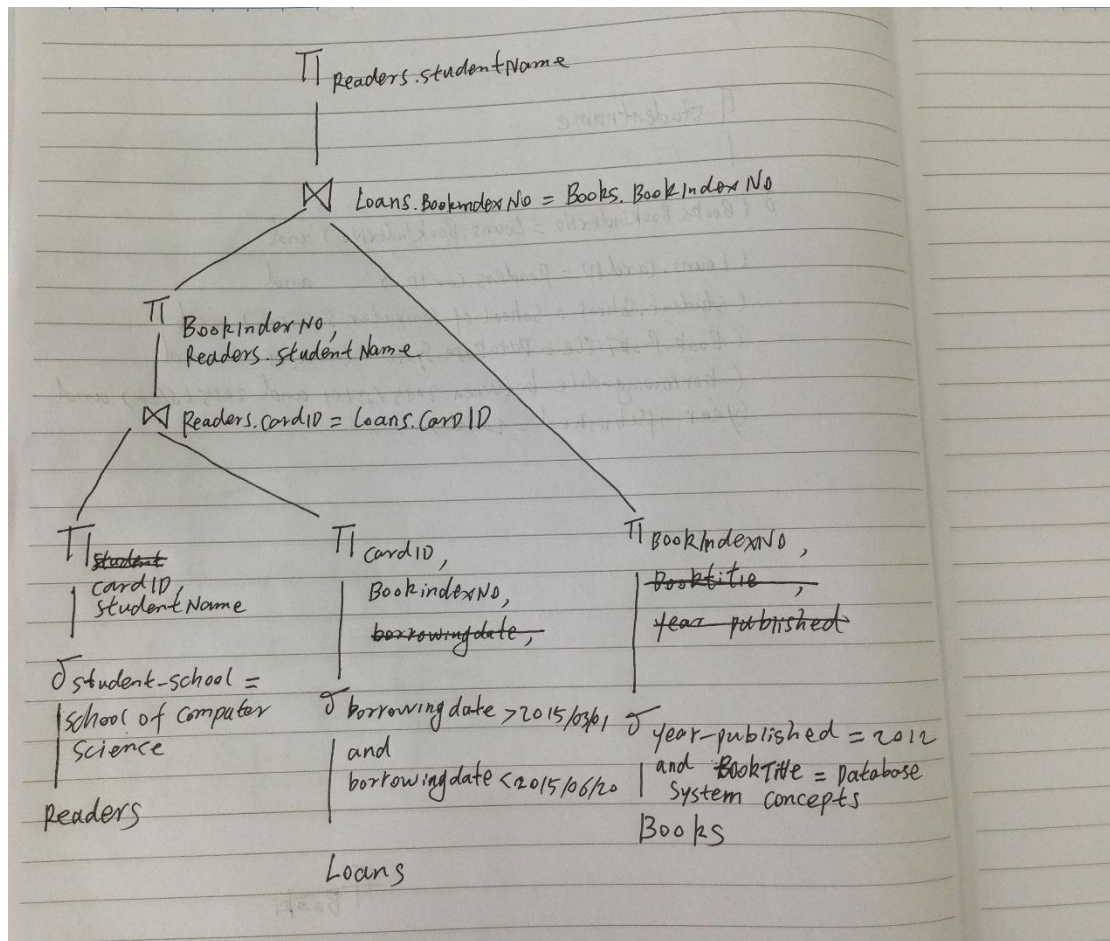
Book.BookTitle= Database System Concepts and

borrowing-date between 2015/03/01 and 2015/06/20 and

year-published=2012

说明：where 子句中的 2 个连接条件，也可以用 from 子句中的 natural join 实现

(2)



3 个选择操作、4 个投影操作、2 个连接操作

4. Consider the **Student-Project** management database given below.

Student (Sno, Sname, class, major, dept_name)

Project (Pno, Pname, requirement, start_time, end_time)

Teacher (Tno, Tname, phone, email, dept_name)

Department (dept_name, building, budget)

Participation (Sno, Pno, Tno, grade)

(1) Give a SQL statement to find some students and list their names and departments that they belong to and list the names of the teachers who guide their projects. It is required that the students' major is 'Big data major', they participate in the projects named as 'Big data analysis and mining', their teachers who guide them are in Computer Science Department, and their grades of projects are more than 85.

```

select  Sname, dept_name, Tname
from    Student, Participation, Project, Teacher
where Student.Sno=Participation.Sno
        and Teacher.Tno=Participation.Tno
        and Project.Pno=Participation.Pno
        and major=' Big data major'
        and Project.Pname=' Big data analysis and mining'
        and Teacher. dept_name=' Computer Science Department'
        and grade>85

```

或者:

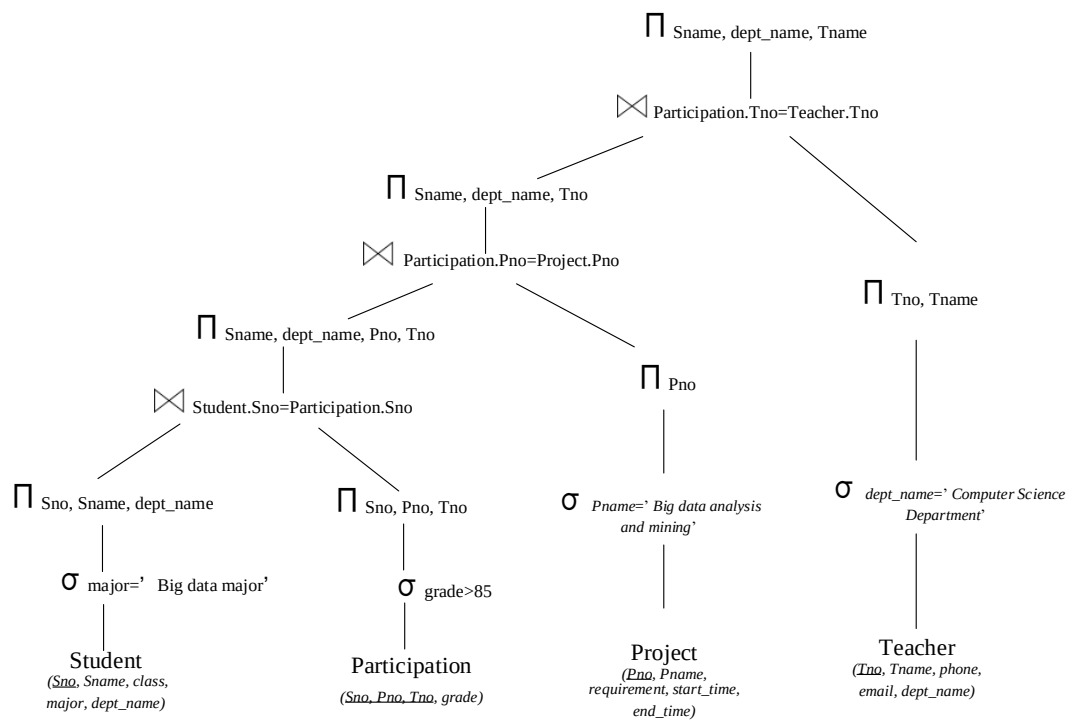
```

select  Sname, major, Tname
from    Student natural join Participation natural join Project natural join Teacher
where major=' Big data major' and Project.Pname=' Big data analysis and mining'
        and Teacher. dept_name=' Computer Science Department' and grade>85

```

说明：4 张表，3 个连接条件，4 个选择条件

(2) For the SQL statement in (1), give an optimized query tree. (7 points)



4 个选择操作，7 个投影操作，3 个连接操作。